

Week 3

Attendance

- In-class? Display student card and tutors will mark you off

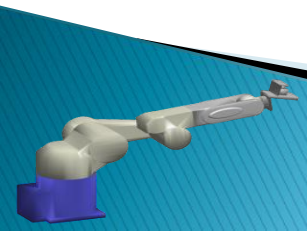
Schedule

- ▶ Quiz 1:
 - Individual (~30 mins)
- ▶ Assignment 1 discussion (15 mins)
- ▶ Review & discussion of Lab 2 (5 mins)
- ▶ Introduction to Lab 3 (5 mins)
- ▶ Collaboratively work on Lab 3 ($\approx$ 2 hours)
- ▶ Prework for Lab 4 (5 mins)



Overview of Week 3: Quiz 1

- ▶ Individual quiz (in-class)
 - worth 5%
 - will start at the same time as the lab
 - approx. 10 points worth of questions
 - no talking or communicating
 - restricted open book (i.e. all canvas material, lab starters & solutions)
- ▶ Re-attempt quiz (after class & only if needed)
 - worth 0%
 - Only required if you get less than 80% for the in-class quiz
 - To be done prior to the next in-class task to unlock that
 - Can only be done outside of class time
 - Can be done alone or collaboratively, but submitted individually
 - Similar number and type of questions as in-class quiz
 - Unlimited attempts allowed

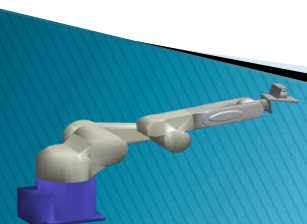


Lab Assignment 1

- ▶ Name: Dual Cobot Assembly task
- ▶ Worth: 20% of subject

Task	Weight	Due
1. Safety Documentation	4%	Marked by a tutor in week 4 or 5 class, then the documentation is submitted to Canvas by 21:00 Wednesday 25th August 2025
2. Real Robot Control from Matlab/Python	2%	Marked off by a tutor in class: Week 5 drop-in lab or Week 5 class or the Week 6 drop-in lab.
3. Online Demo & Code Viva	14%	Week 6 (Wednesday 3rd September 2025) Approximately 20 minutes online during normal class time – bookings required

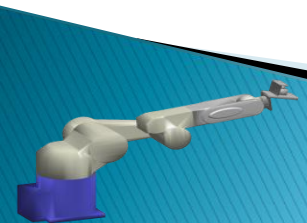
<https://canvas.uts.edu.au/courses/36549/assignments/234275>



Drop-in Lab Sessions:
CB1 1.10.403 in Weeks 4, 5, 6, 10, 11, 12

Timeslots and dates are:

- 9–3pm on 18/8, 25/8, 1/9
- 9–3pm on 9/10, 16/10, 23/10



Pre-work before Week 4 lab

- ▶ Completed Lab 3 exercises (solution available after class)
- ▶ Review “Week 3 – D&H Reference Guide.pdf” on Canvas
- ▶ Ensure you have achieved 80% in Quiz 1
- ▶ Watch the videos in the learning modules on kinematics
- ▶ Attempt Lab 4 exercises
- ▶ Work on Lab Assignment #1 (due week 6)



Note/slides from the textbook

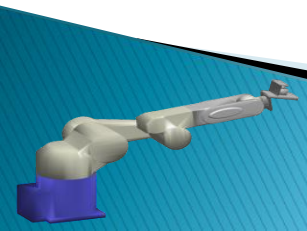
- ▶ Textbook readings for week 3:
 - Section 7.2: “Forward Kinematics”
- ▶ Although it is better to read the textbook, some notes (in slide format) have been summarised below



Forward Kinematics

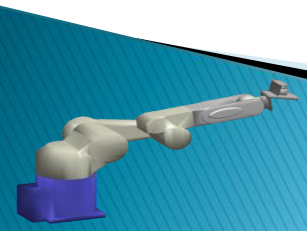
- ▶ The forward kinematics is often expressed in functional form $\xi_E = K(q)$
- ▶ Using homogeneous transformations this is simply the product of the individual link transformation matrices which for an N-axis manipulator is:

$$\xi_E \sim {}^0T_E = {}^0A_1 {}^1A_2 \dots {}^{N-1}T_N$$



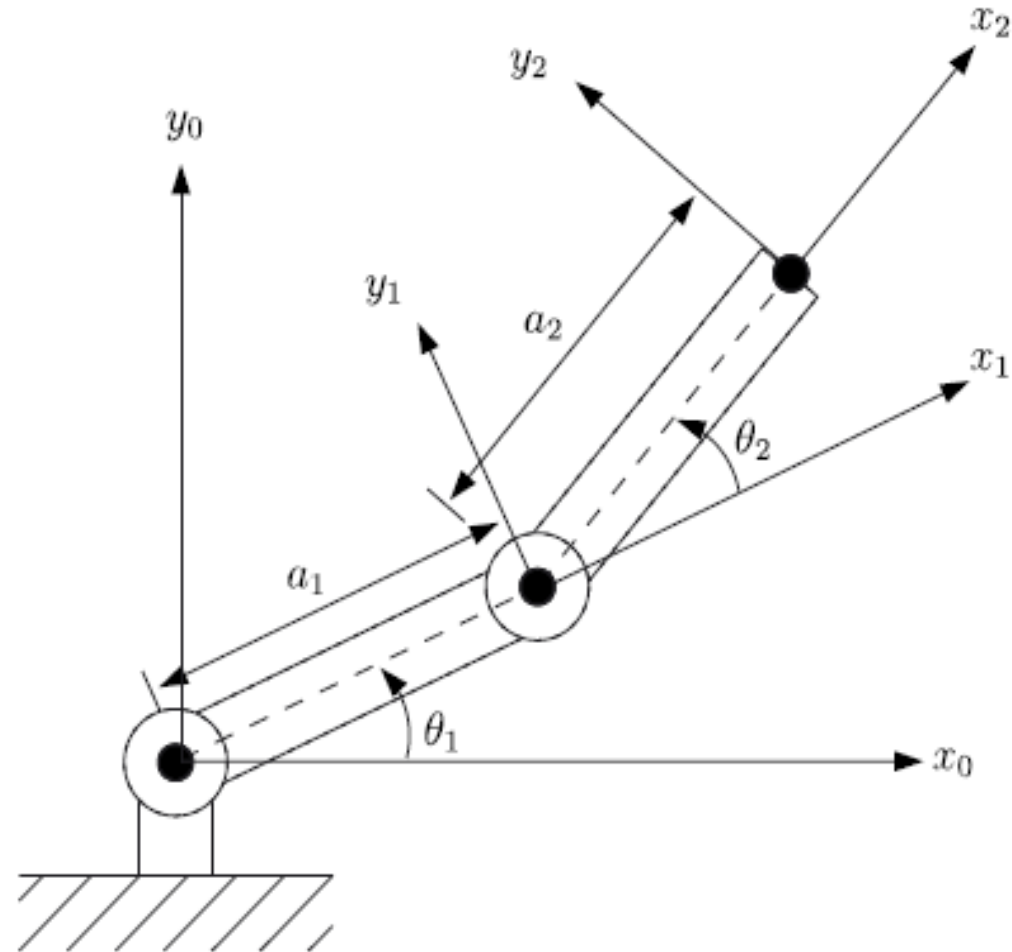
A 2-Link Robot

Link	θ_i	d_i	a_i	α_i	σ_i
1	q_1	0	1	0	0
2	q_2	0	1	0	0



A 2-Link Robot (continued...)

Fig. 7.3. Two-link robot



A 2-Link Robot (continued...)

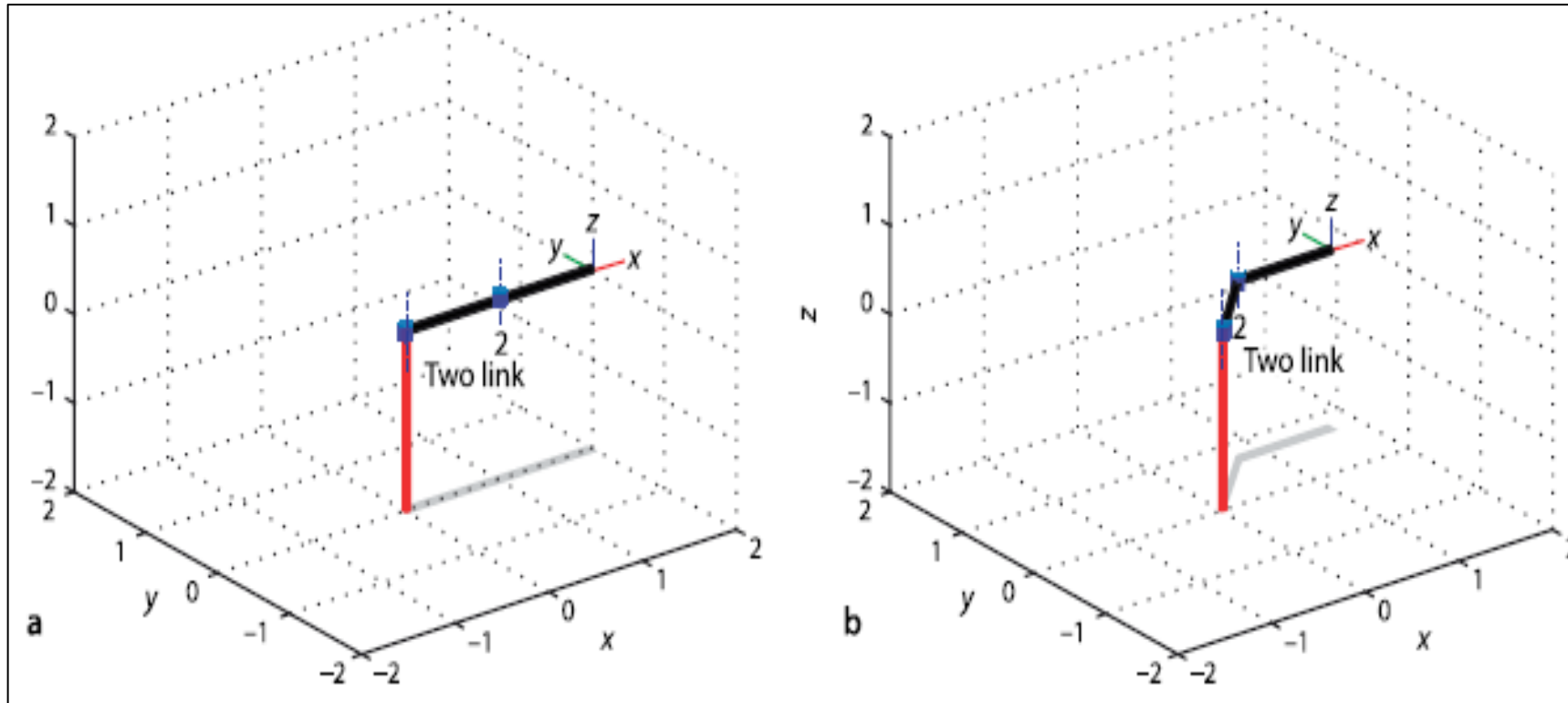
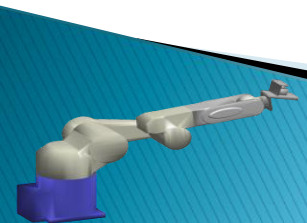


Fig. 7.4. The two-link robot in two different poses, a the pose $(0, 0)$; b the pose $(\frac{\pi}{4}, -\frac{\pi}{4})$. Note the graphical details. Revolute joints are indicated by cylinders and the joint axes are shown as line segments. The final-link coordinate frame and a shadow on the ground are also shown.



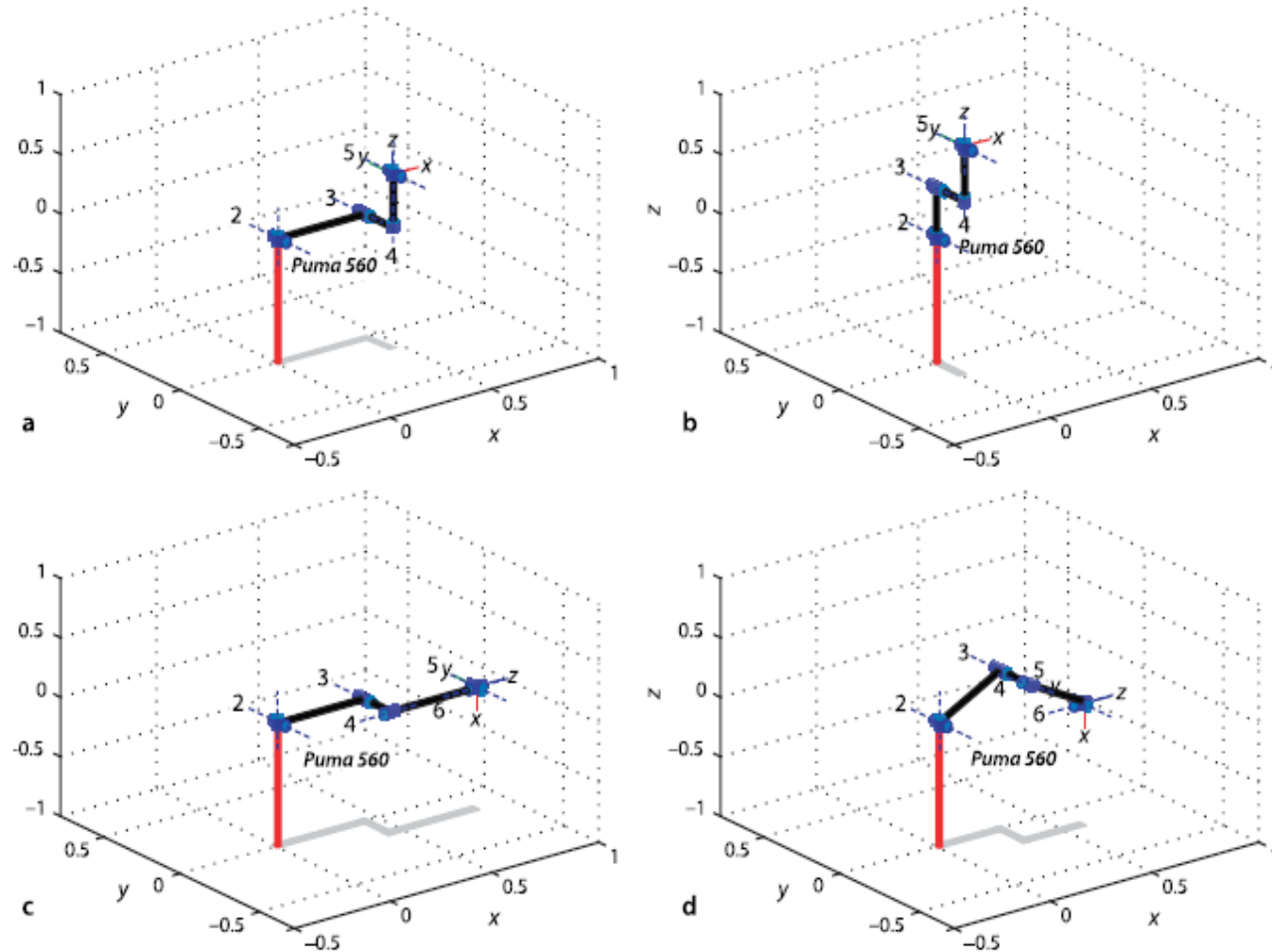
A 6-Axis Robot

- ▶ Truly useful robots have a task space $\mathcal{T} \subset SE(3)$ enabling arbitrary position and attitude of the end-effector.
- ▶ The task space has six spatial degrees of freedom: three translational and three rotational.



A 6-Axis Robot

Fig. 7.5. The Puma robot in 4 different poses.
a *Zero angle*;
b *ready pose*;
c *stretch*;
d *nominal*



Reference

- ▶ Spong MW, Hutchinson S, Vidyasagar M (2006) Robot modeling and control.

